

Additional Binary Trees Problems

Filename: No File Submission

Instructions

This worksheet contains additional practice problems on binary trees. Try to complete each problem on your own before looking at the solutions. The solutions will be provided separately, so make sure to give each problem an honest attempt first. Checking the answer too early defeats the purpose of the practice. As you work, time yourself and do not spend more than 10 minutes on any single problem.

For the following problems, assume the following definitions are provided:

BST Implementation

```
typedef struct BTNode_s {  
    int data;  
    struct BTNode_s *left;  
    struct BTNode_s *right;  
} BTNode;
```

Hint: All of the following problems can be solved recursively. You are not allowed to define any helper functions. Also, note that some questions refer to a binary tree, while others refer to a binary search tree.

TODO 1

Complete the function that returns the number of nodes in a binary tree whose data values are even.

```
int count_even(BTNode *root) {
```

```
}
```

TODO 2

Complete the function that returns the number of leaf nodes in a binary tree. Recall that a leaf node is a node that has no children.

```
int count_leaves(BTNode *root) {
```

```
}
```

TODO 3

Complete the function that returns the number of internal nodes in a binary tree. Recall that an internal node is a node with at least one child.

```
int count_internal(BTNode *root) {
```

```
}
```

TODO 4

Complete the function that returns the sum of all node values in a binary tree.

```
int sum(BTNode *root) {
```

```
}
```

TODO 5

Complete the function that determines whether a binary tree is complete. A binary tree is considered complete if every node has either 0 or 2 children.

```
int is_complete(BTNode *root) {
```

```
}
```

TODO 6

Complete the function that returns 1 if every non-leaf node in a binary tree has a value greater than or equal to the values of its immediate children. Otherwise, return 0.

```
int foo(BTNode *root) {
```

```
}
```

TODO 7

Complete the function that returns the maximum value stored in a binary search tree. You may assume that the tree is not empty.

```
int find_max(BTNode *root) {
```

```
}
```